

Sensory Cravings: A Mixed Reality Installation Enhancing Psychological Experiences through Multisensory Interactions

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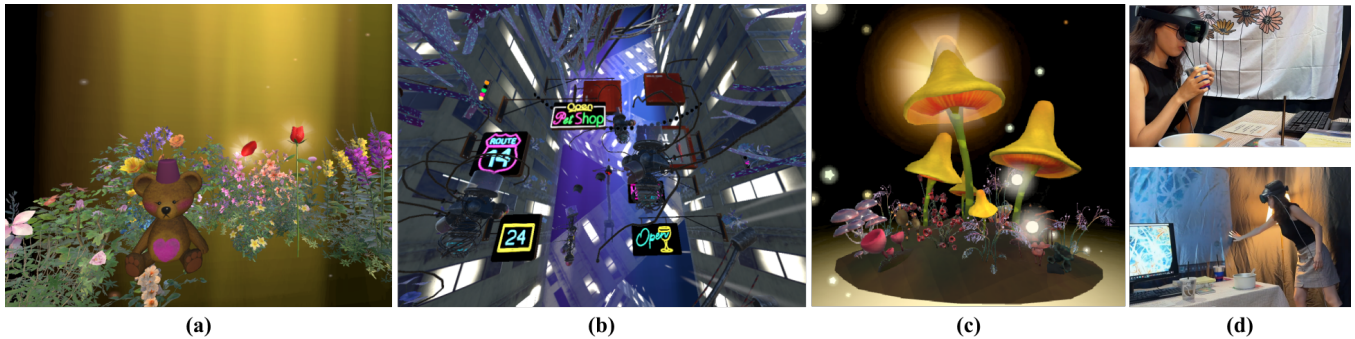


Figure 1: The overview of Sensory Cravings (a) Coffee scene display (b) Alcohol scene display (c) Dessert scene display (d) Multi-sensory experience.

Abstract

Recent advancements in virtual and mixed reality have created opportunities to explore how sensory experiences influence emotional and psychological states. Our installation, Sensory Cravings, leverages mixed reality to create immersive environments simulating the emotional impact of everyday consumables like coffee, alcohol, and desserts. Using Meta's hand-tracking technology and Arduino-based sensor feedback, this work integrates visual, auditory, gustatory, olfactory, and tactile feedback to mitigate modern stress and promote emotional well-being. By incorporating real food choices that guide the virtual experience, the system enhances interactivity and immersion while offering innovative ways to perceive the real world within VR settings, adding to the existing body of work on multisensory integration in virtual cognitive environments.

CCS Concepts

• **Human-centered computing** → *Mixed / augmented reality.*

Keywords

Mixed Reality, Multisensory Integration, Virtual Reality

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1 Introduction

As technological advancements reshape interaction paradigms, immersive technologies are increasingly used not only for entertainment but also as essential tools for understanding and enhancing human emotional and psychological well-being. Recent research has focused on the effects of immersive experiences and multisensory interactions on emotional and psychological states [Morris et al. 2023; Wagener et al. 2022]. For instance, Harley et al. explored the potential of integrating olfactory, tactile, gustatory, auditory, and visual senses within virtual reality, highlighting the use of low-cost non-digital materials and their impact on sensory interaction design [Harley et al. 2018]. Similarly, Kitson et al. proposed a comprehensive framework of design principles and methods, focusing particularly on using scents, sounds, lighting, and physical props to evoke and manipulate profound emotional experiences. [Kitson et al. 2020]. Additionally, the process of data physicalization using candy was examined by Lallemand et al. , emphasizing the importance of multisensory integration in conveying information and enhancing user engagement [Lallemand and Oomen 2022]. However, these studies are predominantly confined to laboratory settings or small-scale explorations with specific groups, lacking empirical research on widespread applications in everyday environments. They do not fully integrate multiple senses into a coherent interaction environment or address how to enhance deeper participant experiences. In

this context, Sensory Cravings utilizes mixed reality to create virtual scenes in everyday environments, integrating multiple sensory modalities to explore how certain food components influence emotional changes and psychological states, enhancing interactivity and immersion through the fusion of physical and virtual elements.

2 Design and Implementation

The experience is set within a 3-by-3 meter area. Participants use a Meta Quest Pro VR headset connected to Unity via Quest Link. They are presented with coffee, wine, and cake. After selecting an item, placing it on the 'confirmation table,' and tasting it, participants put on the headset to enter the corresponding scene.

2.1 Scene Design

To ensure cohesiveness, three scenes are linked through plant elements (Figure 1). Each interactive object provides visual cues via a halo and emits auditory feedback when a participant's hand is detected. The dessert scene is a vibrant garden representing the instant pleasure and allure of sugar. Touching a rose with a golden halo makes it bloom, while interacting with the glowing petals triggers a particle effect of flowers in the sky. Pinching the index finger and thumb together draws a creamy line, transforming the sweetness of the cake into a vivid visual representation. Participants' actions bring the environment to life, promoting sensory engagement and conscious exploration. The alcohol scene presents a dizzying psychedelic world, where organic plant forms and neon lights intermingle with moving biomechanical machines and towering buildings, simulating the enticing dependence, loss of orientation, and sensory overload and distortions caused by alcohol. Participants can also drag a perspective window to explore the subtle boundaries between the VR space and reality. In the coffee scene, touching small mushrooms on the floor produces crisp xylophone sounds of varying pitches, creating a random melody. The richly saturated colors of the plants, along with the light blue glowing currents generated by touching ferns on the wall and giant mushrooms, symbolize the neurologically stimulating and invigorating effects of caffeine (Figure 2). According to Bhumiratana et al.'s research, certain odors evoke similar emotional responses, with dark-roast coffee often linked to positive, high-energy emotions [Bhumiratana et al. 2019]. Future scene iterations will incorporate the scent of dark-roasted coffee beans to create an evocative experience.

2.2 Technical Implementation

The Unity project file includes the Meta SDK integration package required for development on the Meta Quest Pro platform. Scripts attached to objects in the scene enable these objects to float, be positioned vertically, or be placed directly on surfaces scanned and tagged by the headset. These scripts allow the 3D scene to dynamically adapt to the room layout, thereby enhancing immersion. For example, flowers or clusters of mushrooms are displayed on surfaces tagged as 'floor.' In the alcohol scene, the headset camera, through the implemented scripts, detects the gesture of pinching the thumb and middle finger together, switching the environmental ambience and providing audio feedback. An RFID sensing module identifies electronic tags beneath the three items, with Arduino transmitting the corresponding UID number to Unity via a script

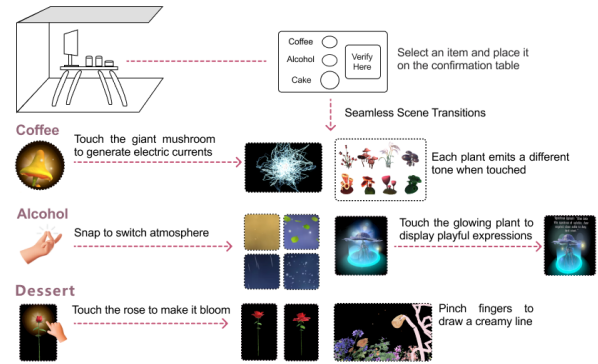


Figure 2: Sensory Cravings Interaction Flowchart

that applies across all scenes, enabling seamless transitions between scenes.

3 Conclusion

This work uses mixed reality technology to create digital spaces for emotional release, grounded in real-world settings. By integrating multisensory interactions, it aims to alleviate stress and promote mental well-being. Virtual representations of emotions elicited by food, combined with gesture tracking and sensory feedback, provide a quality interactive experience. User feedback indicates its effectiveness in delivering emotional depth and interaction, showing its potential as a psychological engagement tool.

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